

Mekonnen Tesfazien

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SUMMARY

Software Developer skilled in designing scalable solutions and delivering high quality results through strong technical expertise and problem solving and communication skills abilities. Known for leveraging modern tools and methodologies to create impactful, user-centric applications.

WORK EXPERIENCE

Researcher

12/2023 – 01/2026 | San Francisco, CA

San Francisco State University

- Conducted research on Large Language Models (LLMs), focusing on advanced applications in natural language processing (NLP) such as UI understanding, QA, or multimodal grounding.
- Designed and fine-tuned transformer-based models using PyTorch, TensorFlow, and Hugging Face Transformers, achieving improved performance on [dataset/task—add metric if possible].
- Trained models on an HPC cluster using distributed training (DDP), optimizing GPU utilization and reducing training time by 90%.
- Developed and managed scalable training pipelines using Slurm and Bash scripting for job scheduling and resource allocation.
- Performed hyperparameter tuning and loss function optimization using IoU, SmoothL1 and L2 to improve model accuracy in bounding box prediction tasks.
- Collaborated with HPC engineers to debug cluster-level issues, monitor GPU performance, and ensure efficient resource usage.

Software Developer

11/2020 – 09/2022 | Asmara, Eritrea

CO - Bounce

- Delivered a responsive full-stack UI by employing JavaScript, ES6, React, and Redux, significantly enhancing client interaction and overall user experience
- Redesigned project with UX designers, using React, Java, Spring Boot, and Hibernate, boosting user engagement by 18% and use caching to increase load time.
- Deployed applications using Docker for containerization and efficient deployment.
- Performed unit, integration, and end-to-end tests with Jest, Mocha, and React Testing Library.
- Contributed to comprehensive documentation using Confluence, ensuring effective knowledge sharing within the team, customer service and enhancing overall project understanding.

Software Developer

12/2018 – 11/2019 | Asmara, Eritrea

ERITEL

- Developed a complex e-commerce platform with React, and Java, featuring a product catalogue, shopping cart, and user authentication with TCP/IP stack.
- Designed and deployed applications using Java, React, Spring Framework, and Hibernate, improving reliability and code quality with software development tools and Agile practices.
- Deployed and hosted Java applications using AWS Amplify, S3, and CloudFront and maintain JSON API.
- Developed e-commerce solutions with Angular and incorporating Bash scripts for deployment automation.
- Supported embedded system integration for switching and telemetry units programmed in C/C++.
- Utilized Docker for containerization and deployment and collaborated with UX/UI team for appealing designs and seamless user experiences.

Electrical Engineer

09/2016 – 08/2018 | Asmara, Eritrea

Keih Bahri Food Production Company

- Integrated and calibrated sensors used in monitoring temperature, flow, and production parameters, ensuring accurate data collection and system reliability
- Successfully managed multiple projects, consistently meeting 100% of aggressive timelines and delivering all milestones on schedule.

- Collaborated with different teams and department, providing clear communication and technical support to ensure project success.
- Utilized electronic AutoCAD for effective PCB design and layout while using C/C++ for embedded system configuration.
- Leveraged Linux for monitoring and testing environments.
- Designed and implemented an automated testing system for electrical components, reducing testing time by 50% and increasing overall production efficiency by 20% using MATLAB and LabVIEW.
- Collaborated with cross-functional teams to coordinate electrical work, resolve system issues, and meet production deadlines
- Supported preparation of technical documentation, reports, and maintenance records for engineering and operations teams

EDUCATION

Master of Science in Computer Science

San Francisco State University

GPA: 3.8/4.0

09/2023 – 01/2026 | San Francisco, CA

Bachelor of Science in Electrical and Electronics Engineering

Eritrea Institute Of Technology

GPA: 3.9/4.0

09/2011 – 06/2016 | Mai Nefhi, Eritrea

SKILLS

Frontend & UI Tools: JavaScript, TypeScript, React, Redux, Next.js, HTML, CSS, SASS, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, Python, Java, Spring Boot, Hibernate

Databases: SQL (MySQL, Postgres, SQLite), NoSQL (MongoDB)

Cloud & Deployment: AWS (Lambda, S3, EC2), Docker, Kubernetes, CI/CD pipelines

Testing & Automation: Jest, Mocha, Selenium, TestCafe, TDD, E2E Testing

Development Practices: OOP, Agile, Scrum, SDLC, Microservices architecture, Software Architecture, Saas

Professional Skills: Technical Communication & Documentation, Cross-functional Collaboration, Problem Solving

Other: Technical support, troubleshooting, DevOps integration, process improvement, Jira

PROJECTS

[UI Element Detection with Vision Language Models](#)

Python, PyTorch, Hugging Face, BLIP-2, QLoRA,

[Custom File System](#)

C, POSIX system calls, bio library, multithreading.

[E-Commerce Website](#)

React, Redux, NodeJS, user authentication, Stripe API payments.

[AI Resume Builder](#)

React, Node.js, ChatGPT API integration.